

Decoding Stem Cell Growth Mechanisms with Physics-Informed Data Augmentation

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Introduction

Stem Cells

Pluripotent stem cells can yield diverse therapeutics [1], yet large-scale manufacturing remains constrained by resource-intensive protocols and low experimental throughput [2]. Glucose and lactate dynamics serve as accessible proxies for pluripotency and growth [8,9]. Thus, this study will **leverage metabolic indicators to guide process optimization with ML**.

Machine Learning

ML preserves bioprocess fidelity during scale-up [3] and maps process parameters to CQAs [4], yet its interpretability and robustness decline with limited data [5]. Physics-informed digital twins integrate mechanistic priors directly into ML architectures [6,7]. We evaluate how **architectural constraints modulate both accuracy and insight quality**.

Step 1: Experimental Data Collection

Process Variables $y = [X_V, X_T, X_{NAN+}, GLC, LAC]^T$

Every 24 hours, we measured **five variables**:

- Viable, total, and pluripotent (NANOG+ [5]) cell densities
 - Glucose and lactate concentrations
- Measurements spanned **eight conditions**:
- Two cell lines (hESCs and hiPSCs)
 - Four glucose feeds (1.0, 5.0, 17.5, 20.0 mM)

Step 2: Digital Twin Optimization

No Prior Knowledge
(Limited assumptions)

First Principles
(Strong assumptions)

Neural ODE (nodeAM)

10255 NN Parameters
Mostly statistical,
assume scaling factor
from variance of data

$$\frac{dy_i}{dt} = \sigma^2(y_i)F_{NN}(y, p)$$

Hybrid Digital (hdAM)

115 NN Parameters
Assume stoichiometry,
maximum rates,
dependency of VCD

$$\frac{dy_i}{dt} = \mu_{max,i} \nu_i X_v F_{NN}(y, p)$$

First Principle (fpAM)

16 Physical Parameters
Mostly physical, modified
mAB Monod kinetic
model of cell growth

$$\frac{dX_v}{dt} = (\mu_{max} f_{lim} f_{inh} - \mu_d) X_v$$

Parameter Estimation

Parameterize by minimizing normalized trajectory error plus CQA-matching penalty.

$$\arg \min_p \mathcal{L}(y, p) = \sum_{j=1}^k \sum_{i=1}^n \frac{(\hat{y}_{ij}(p) - y_{ij})^2}{\sigma^2(y_i)} + \frac{(CQA(\hat{y}) - CQA(y))^2}{\sigma^2(CQA(y))}$$

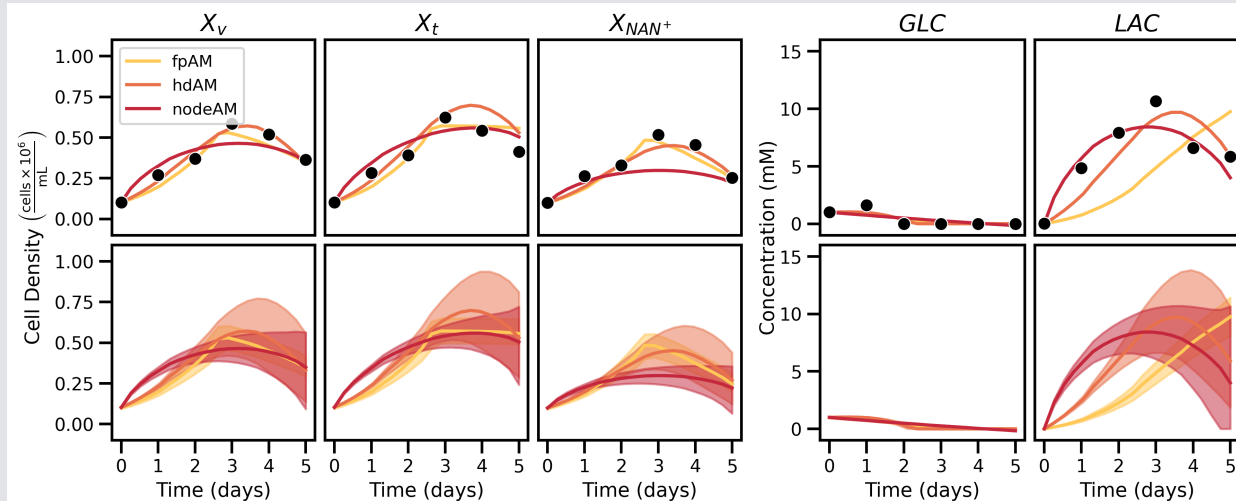


Figure 1. Optimized augmentation models overlaid with experimental data points (top). Shaded regions represent the 95% confidence intervals of the augmented data sets from each model (bottom) for iPSCs with 1.0 mM feed.

Under low-glucose conditions (1.0 and 5.0 mM), the hdAM and nodeAM significantly outperformed the fpAM (p=0.041 and p=0.009, respectively). All models can capture the shapes of experimental data, and the hdAM achieved the lowest loss values overall despite having fewer parameters than the nodeAM.

Step 3: Data Augmentation and Regression

Augmentation Procedure

Digital twins were integrated under perturbed conditions (Fig. 1) generated by Gaussian sampling of the initial conditions and specified aspects of each model:

Augmentation Model	Means	Variance
First Principles (fpAM)	7 of the 16 optimal constants	5%
Hybrid Digital (hdAM)	Maximum growth rates	1%
Neural ODE (nodeAM)	Variance scaling factor	2.5%

Regression and Feature Importance

Multiway partial least squares regression was applied to augmented glucose and lactate concentration datasets (X [N × 2 × 20]) to predict harvest viable cell density and pluripotency (Y [N × 2]) using four latent variables. Subsequently, a Monte Carlo-based VIP analysis quantified each time point's likelihood of significance (VIP > 1), and the resulting normalized importance scores are displayed as heat maps (Fig. 3).

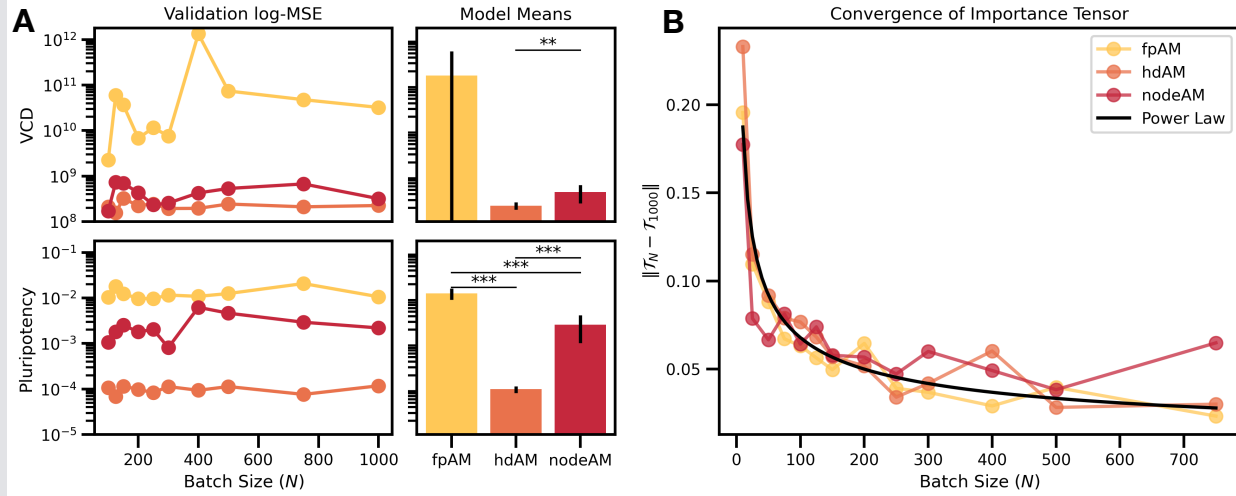


Figure 2. Validation log-MSE for the CQAs of viable cell density and pluripotency with the mean log-MSE for each model (A) and power law convergence of the feature importance tensor for each augmentation model (B).

Validation MSE stays relatively constant with augmented sample size (A), but with significant differences between the correlation strengths between glycolysis and both CQAs (from best to worst: hdAM, nodeAM, fpAM). The feature importances of the regression models converged following a power law with batch size (B).

Step 4: Biological Trend Identification

FDA-ANOVA

Use functional data analysis (FDA) ANOVA on temporal VIP trajectories as functions of importance over time to test how experimental factors (cell line, model architecture, etc.) significantly alter learned feature-importance patterns.

Biological Trend	fpAM	hdAM	nodeAM
Similarity across cell lines [48]	p=0.636	p=0.780	p=0.837
Different functional roles of glucose and lactate	p=0.012	p=0.012	p=0.1075
Glucose Imp increases with feed concentration [9,44]			
Importance changes align with rate shifts [45]			
Lactate Imp increases with accumulation [43]			
Lactate Imp overall higher than glucose [44]			

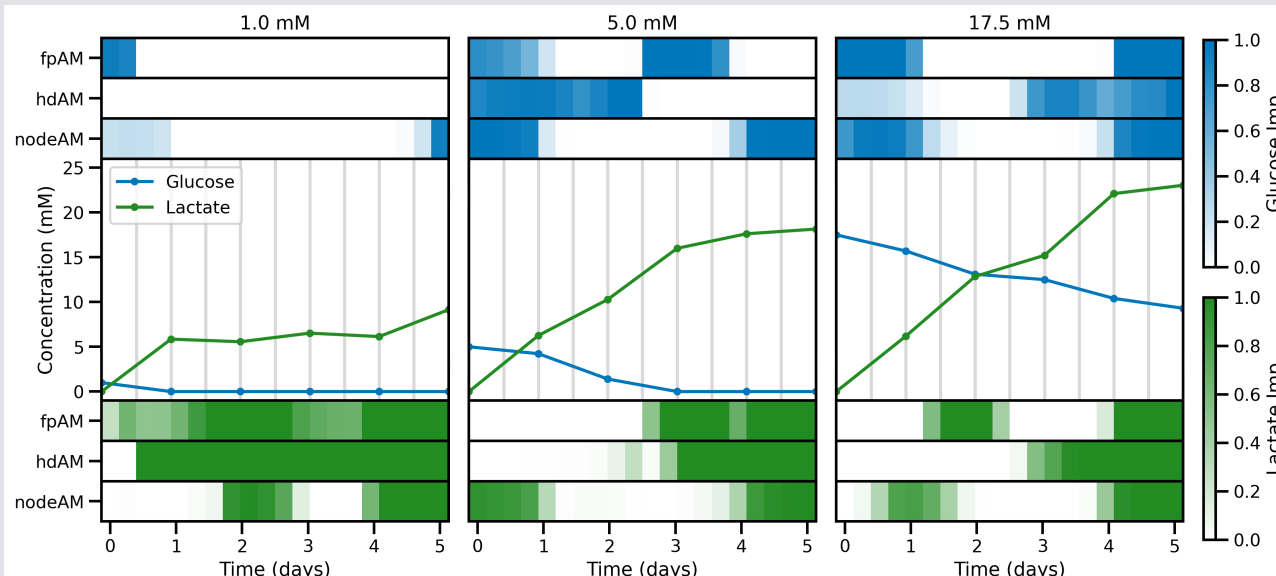


Figure 3. Key temporal windows measured by feature importance of glucose and lactate concentrations in the machine learning models correlating to the CQAs (percentage of pluripotency cells expressing NANOG and viable cell density) at various glucose feed concentrations for iPSCs.

Discussion

Performance Metric	fpAM	hdAM	nodeAM
Fit to data	3	1	2
Correlation strength	3	1	2
Biological interpretability	2	1	3

Architecture Comparison

By embedding stoichiometric and viable-cell dependencies in the hdAM, this model outperformed the mechanistic fpAM and the unconstrained nodeAM. These mechanistic constraints act as regularizers, reducing data volume and training load while preserving model flexibility, consistent with the advantages of hybrid digital bioprocess models [38,39] and the principle of balancing soft constraints with statistical learning [18]. This demonstrates how targeted architectural priors can optimize predictive accuracy and interpretability in complex temporal systems.

Future Work and Conclusions

Application 1: Stem Cells

The use of glucose and lactate as features extends naturally to include additional molecular indicators of stem cell growth (e.g., fibroblast growth factors [48] and ammonia [49]). By systematically comparing architectures akin to the hdAM, we can iteratively refine model constraints and uncover how specific mechanistic assumptions translate into learnable insights about stem cell metabolism. This modular framework enables ranking and integration of new biomarkers, guiding targeted experimental validation and accelerated optimization of culture protocols.

Application 2: General Systems

This framework readily extends to any coupled temporal system by defining relevant time-series variables, constructing augmentation models spanning assumption-based to statistical extremes, and leveraging feature-importance analysis to reveal critical dynamic windows. Thus, practitioners can systematically optimize model architecture and interpretability across diverse domains.

Acknowledgement

The research is partially supported by the National Science Foundation (NSF) grant CBET-2326510.

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